



## Setup & Configuration

### Configuration Options

**theme**: Customize colors, spacing, fonts, breakpoints, etc.

#### Example:

```
theme: {
  extend: {
    colors: {
      'custom-blue': '#123456',
    }
  }
}
```

**plugins**: Add custom styles and functionality.

#### Example:

```
plugins: [
  require('@tailwindcss/forms'),
]
```

**variants**: Control which variants are enabled for each utility.

(Deprecated in Tailwind CSS v3.0+; use the **content** configuration instead)

**prefix**: Add a prefix to all Tailwind CSS class names to avoid conflicts.

#### Example:

```
prefix: 'tw-'
```

**corePlugins**: Enable or disable specific Tailwind CSS core plugins.

#### Example:

```
corePlugins: {
  container: false,
}
```

**darkMode**: Enable dark mode based on prefers-color-scheme or class name.

#### Example:

```
darkMode: 'class'
```

**content**: Specify the files to scan for usage of Tailwind CSS classes.

#### Example:

```
content: [
  './public/**/*.html',
  './src/**/*.js',
  './src/**/*.jsx',
  './src/**/*.ts',
  './src/**/*.tsx',
]
```

**safelist**: Specify classes that should not be removed during production builds.

#### Example:

```
safelist: [
  'bg-red-500',
  'text-green-200',
  'lg:bg-blue-400'
]
```

**blocklist**: Specify classes that should always be removed during production builds.

#### Example:

```
blocklist: [
  'bg-red-500',
  'text-green-200',
  'lg:bg-blue-400'
]
```

### Typography Styles

|                 |                                           |
|-----------------|-------------------------------------------|
| font-sans       | Apply a sans-serif font family.           |
| font-serif      | Apply a serif font family.                |
| font-mono       | Apply a monospace font family.            |
| text-xs         | Extra small text size ( 0.75rem / 12px ). |
| text-sm         | Small text size ( 0.875rem / 14px ).      |
| text-base       | Base text size ( 1rem / 16px ).           |
| text-lg         | Large text size ( 1.125rem / 18px ).      |
| text-xl         | Extra large text size ( 1.25rem / 20px ). |
| font-thin       | Font weight 100.                          |
| font-extralight | Font weight 200.                          |
| font-light      | Font weight 300.                          |
| font-normal     | Font weight 400 (normal).                 |
| font-medium     | Font weight 500 (medium).                 |
| font-semibold   | Font weight 600 (semi-bold).              |
| font-bold       | Font weight 700 (bold).                   |
| font-extrabold  | Font weight 800 (extra-bold).             |
| font-black      | Font weight 900 (black).                  |
| italic          | Set font style to italic.                 |
| not-italic      | Set font style to normal (non-italic).    |
| uppercase       | Transform text to uppercase.              |
| lowercase       | Transform text to lowercase.              |
| capitalize      | Transform text to capitalize.             |

## Core Concepts & Utilities

### Utility-First Fundamentals

Tailwind CSS uses a utility-first approach, meaning you compose styles by applying pre-defined utility classes directly in your HTML.

#### Example:

```
<button class="bg-blue-500 hover:bg-blue-700 text-white font-  
bold py-2 px-4 rounded">  
  Click me  
</button>
```

**Responsive Design:** Use breakpoint prefixes like `sm:`, `md:`, `lg:`, `xl:`, and `2xl:` to apply styles at different screen sizes.

#### Example:

```
<div class="md:flex">  
  <!-- ... -->  
</div>
```

**Hover, Focus, and Other States:** Use prefixes like `hover:`, `focus:`, `active:`, `disabled:` to style different element states.

#### Example:

```
<button class="hover:text-green-500 focus:outline-none  
focus:ring-2 focus:ring-green-500">  
  Hover me  
</button>
```

**Dark Mode:** Use the `dark:` prefix to style elements in dark mode (requires dark mode to be enabled in your `tailwind.config.js`).

#### Example:

```
<div class="bg-white dark:bg-gray-800">  
  <!-- ... -->  
</div>
```

## Customization & Directives

### Adding Custom Styles

While Tailwind encourages utility classes, you can add custom CSS:

- Using `@apply`:** Apply existing Tailwind utility classes within your CSS.

```
.btn-primary {  
  @apply bg-blue-500 hover:bg-blue-700 text-white font-bold  
  py-2 px-4 rounded;  
}
```
- Directly in CSS:** Write standard CSS rules in your CSS files. Tailwind will handle the rest.

```
.my-custom-class {  
  color: purple;  
  font-size: 1.2rem;  
}
```

### Essential Utilities

|                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                      |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Typography:</b> <ul style="list-style-type: none"><li><code>font-{family}</code></li><li><code>text-{size}</code></li><li><code>font-{weight}</code></li><li><code>italic</code>, <code>not-italic</code></li><li><code>underline</code>, <code>line-through</code></li><li><code>text-{color}</code></li></ul>                  | <b>Background:</b> <ul style="list-style-type: none"><li><code>bg-{color}</code></li><li><code>bg-{image}</code></li><li><code>bg-{position}</code></li><li><code>bg-{size}</code></li><li><code>bg-repeat</code>, <code>bg-no-repeat</code></li></ul>                                                                               |
| <b>Spacing (Margin &amp; Padding):</b> <ul style="list-style-type: none"><li><code>m{t b l r x y}-{size}</code></li><li><code>p{t b l r x y}-{size}</code></li></ul>                                                                                                                                                                | <b>Flexbox &amp; Grid:</b> <ul style="list-style-type: none"><li><code>flex</code>, <code>inline-flex</code></li><li><code>grid</code>, <code>inline-grid</code></li><li><code>flex-{row col}</code></li><li><code>justify-{content}</code></li><li><code>items-{alignment}</code></li><li><code>grid-cols-{number}</code></li></ul> |
| <b>Border:</b> <ul style="list-style-type: none"><li><code>border</code></li><li><code>border-{width}</code></li><li><code>border-{color}</code></li><li><code>rounded</code>, <code>rounded-{size}</code></li></ul>                                                                                                                | <b>Positioning:</b> <ul style="list-style-type: none"><li><code>static</code>, <code>relative</code>, <code>absolute</code>, <code>fixed</code>, <code>sticky</code></li><li><code>top-{value}</code>, <code>right-{value}</code>, <code>bottom-{value}</code>, <code>left-{value}</code></li><li><code>z-{index}</code></li></ul>   |
| <b>Display:</b> <ul style="list-style-type: none"><li><code>block</code>, <code>inline-block</code></li><li><code>inline</code></li><li><code>flex</code>, <code>inline-flex</code></li><li><code>grid</code>, <code>inline-grid</code></li><li><code>table</code>, <code>inline-table</code></li><li><code>hidden</code></li></ul> | <b>Effects:</b> <ul style="list-style-type: none"><li><code>shadow</code>, <code>shadow-{size}</code></li><li><code>opacity-{value}</code></li><li><code>blur</code>, <code>blur-{size}</code></li><li><code>transition</code>, <code>duration-{time}</code></li><li><code>transform</code>, <code>scale-{value}</code></li></ul>    |

Useful Directives

|                                                                                                                                                            |                                                                                                                                                                                                      |
|------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>@tailwind</b> : Injects Tailwind's base, components, and utilities styles.</p> <pre>@tailwind base; @tailwind components; @tailwind utilities;</pre> | <p><b>@layer</b> : Organize your custom CSS into layers (base, components, utilities) for better control over specificity.</p> <pre>@layer components {   .btn-primary {     @apply ...;   } }</pre> |
| <p><b>@apply</b> : Inline existing utility classes into custom CSS rules.</p> <pre>.my-element {   @apply flex items-center; }</pre>                       | <p><b>@config</b> : Specify a custom Tailwind config file (rarely needed).</p> <pre>@config 'path/to/my/tailwind.config.js';</pre>                                                                   |
| <p><b>@variants</b> : (Deprecated) - In Tailwind v2, used to generate variants for custom CSS. Superseded by the <b>content</b> configuration.</p>         |                                                                                                                                                                                                      |

AI Tools & Resources

AI-Powered Tailwind Tools

Several AI tools can help you generate Tailwind CSS code, including:

- TeleportHQ:** AI-powered design-to-code platform that supports Tailwind CSS.
- Locofy.ai:** Convert designs to responsive code, including Tailwind CSS.
- Galytix:** Use AI to generate Tailwind code from Figma designs.
- Windframe:** Quickly prototype website sections with AI-generated Tailwind CSS code.

Official Resources

- Tailwind CSS Documentation:** [tailwindcss.com](https://tailwindcss.com)
- Tailwind UI:** [tailwindui.com](https://tailwindui.com) - Component library built with Tailwind CSS.

Community Resources

- Tailwind CSS Discord:** Active community for support and discussions.
- Tailwind Weekly:** Newsletter with tips, tricks, and resources.
- tailwindcss-plugins:** A repository of useful Tailwind CSS plugins.
- Various online tutorials, blog posts, and videos.**